Analyzing an Unexplored Layer of Digital Audio Forensics with Spectrogram-Based Steganography and Detection

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***Abstract*— Audio evidence analyzed in digital forensics is often examined with metadata extraction, waveform inspection, and audio processing. However, audio can also be represented in visual forms, specifically as spectrograms, a frequency over time visualization depicting patterns in pitch of an audio file. Spectrograms are underutilized in the world of digital forensics despite its ability to hide and manipulate information within audio evidence. This paper presents a practical examination into spectrogram-based steganography and detection, showcasing how text and images can be intentionally embedded into audio files to hide information without altering traditionally examined metadata. Based in Python, the system is developed to embed text or an image into an audio file using its spectrogram. Additionally, the system can detect possible text and image manipulation of an audio file's spectrogram. The experiment shows that spectrogram analysis can expose hidden information that would typically evade traditional audio examination techniques. This work highlights the spectrograms as an underutilized layer of digital audio forensics that would benefit modern digital forensic investigations.**

Keywords—Digital Forensics, audio forensics, spectrogram, steganography, detection, signal processing

I. Introduction

Digital audio is an increasingly important form of evidence used in cybercrime investigations, criminal proceedings, and other legal hearings. Audio data can serve as proof for communications, criminal activity, and authentication of evidence, where in cases it is utilized properly, can be the critical point in an investigative case. Therefore, it is crucial to ensure that related audio data in a case has been thoroughly examined and analyzed, ensuring the integrity and authenticity of data.

Traditionally, digital audio analysis remains focused audio processing, metadata analysis, or simply listening to the audio and interpreting it. As a result, some alternate representations of audio evidence can get overlooked, including visual representations of audio. Typically, audio evidence is examined at a high-level representation, focusing on the audio content and the corresponding metadata of the audio. While this may be great for authenticating audio evidence for use in digital forensic investigations, it may skip over an underexamined layer of audio evidence: an audio's visual representation.

Audio can be visualized into various representations, depicting amplitude, frequency, and loudness. Most commonly shown as waveforms, which show amplitude over time, they can also be represented as spectrograms, which show frequency (pitch) over time. This visualization is not typically considered in forensic settings, rather it is seen as a visualization technique for audio processing and diagnostics. As a result, current digital forensic software will omit this visualization from audio files. This creates an opportunity for adversaries and criminals to hide messages and manipulate audio files that can evade detection from traditional digital audio examination techniques. In turn, malicious audio evidence may be incorrectly classified as benign and aid in a criminal's favor.

A criminal sees this as a point of attack, a way to encode information into an audio file that may not be apparent in metadata or waveforms. Though spectrograms are primarily used to visualize the pitch of an audio over time, it can reveal important information of audio manipulation. Splicing, adding, or deleting sections of audio can be shown, but it can also hold hidden information. To combat this, the paper presents a solution that will investigate spectrogram-based steganography, concealing information and data within a seemingly harmless file, and explore detection strategies for altered audio files.

The proposed system will create a practical approach to investigating and analyzing spectrograms and their use in digital forensics. Based in Python, this framework is designed to intentionally embed visual information, particularly text and images, into the spectrogram of an audio file, optionally offering capabilities to merge with an existing audio file without modifying commonly examined audio metadata. This function will emphasize the importance of spectrogram analysis in digital forensics as an outlet for malicious intent. Furthermore, the solution will also a spectrogram analysis feature that will apply detection techniques to classify if a specific audio file has been encoded with a hidden message. Altogether, the proposed solution acts as a practical investigation for digital forensic practitioners to understand, detect, and visualize hidden information embedded in an audio's frequency-domain.

II. Related Works

Traditional audio analyzation methods are typically focused within specific tasks involved in audio engineering and digital signal processing. Maher's *Overview of Audio Forensics* [1] provides a comprehensive and foundational understanding of audio forensic methodologies. Core investigative tasks rely on authenticating the integrity of audio

files, detecting enhancements and manipulations, and interpreting the relevance as it applies in a legal and investigative context. According to Maher, the use of spectral analysis tools, particularly spectrograms, is an essential part of manipulation detection and visualizing abnormalities in an audio.

Spectrogram analysis an expose artifacts of manipulation through changes of spectral texture caused by splicing or insertion. It can be used as a diagnostic visualization to find deviations in frequency from expected pitches and remains an important authentication tool for validating audio files from alterations. Maher emphasizes that with modern audio editing techniques, manipulated recordings can be deceitful and appear benign in a waveform and metadata level. A practitioner must be able to understand and recognize various audio frequency patterns to differentiate natural audio from altered content. This raises the importance of spectrogram analysis, as frequency information is not as easily preservable. However, the paper does not consider the possibility of using spectrograms as a medium to encode hidden information.

In this context, the paper will extend Maher's research of spectrogram usage. As an important diagnostic tool for audio engineering and signal processing, it should also be considered as a tool for adversarial usage given its ability to bypass metadata and waveform analysis.

Another survey led by Stamm et al. examined various multimedia forensic techniques from the earlier years. Particularly, the researchers claim that most forensic analysis must indulge in deeper analysis techniques to understand the processing history of various media [2]. Their work categorizes forensic challenges in digital evidence and highlights the ability to detect traces of manipulation through signal analysis. Stamm et al. draw attention to the fact that a thorough analysis must involve identifying subtle discrepancies in signal representations. Presented spectrogram analysis techniques can be adapted from their research to develop an accurate system for classifying manipulated audio data.

In their research, Stamm et al. introduces a range of analysis techniques that can be used in audio forensics. They propose frameworks that interpret unnatural audio energies in signal processing. Large concentrations of frequencies and large variance are indicative of manipulation and show an inconsistent structure in a spectrogram. Further analysis can be done by interpreting the visualizations from spectral analysis. Common signs of manipulation include abrupt transitions and structured geometrics, for example, a square shaped frequency range in a spectrogram. These techniques can be adapted and applied to the project, automating the detection process of audio files.

Other recent studies showcase the emergence and effectiveness of spectrograms in research and application. Nguyen-Duc et al. present a comprehensive study of audio models for deepfake singing. As it presents, more recent techniques in singing voice synthesis produces higher quality, artificially produced singing voices [3]. This poses a challenge that stems from the misuse of deepfake singing technology, where deepfake detection is harder than before.

Their findings articulate that models involving spectrogram analysis outperformed the use of waveform-based models. Because spectrograms also contain information about frequency-time patterns, harmonic irregularities, and spectral artifacts that are discernable in waveform representations, utilizing spectrogram-based models are more accurate and efficient. The reliance on spectrograms in deep learning applications has direct implications for digital forensics.

Adopting spectrograms in digital forensics can prove to be more beneficial than traditional methods. Representing audio evidence in various dimensions, like in a two-dimensional representation like a spectrogram, deeper features of the evidence are revealed. Specifically, spectral features introduced through manipulation and other anomalies are emphasized. The shift towards spectrograms usage can show that spectrograms are no longer primarily used for diagnostic and processing information, rather that they are also an important forensic tool that can reveal essential information of a digital audio file.

III. Motivation

Digital audio forensics has been rooted in traditional analysis methods that focus on high-level analysis techniques, including waveform and metadata analysis. While effective in manipulation detection, these approaches can overlook certain aspects of audio information not explicitly stated or represented. Spectrograms offer a two-dimensional representation of audio files that can prove to be beneficial in manipulation detection by analyzing the various frequency-time properties of the file. Despite the benefits of spectrogram analysis, it is often overlooked in digital forensic investigations.

The motivation behind this work is to highlight the importance of spectrograms in digital forensics as an underexamined layer of audio evidence. Because spectrograms reveal information not explicitly prevalent in waveform and metadata examinations, it poses as an optimal medium for people with malicious intent. This solution will explore how it can easily be exploited to hide information.

Furthermore, the use of spectrograms in digital forensics is still fairly limited. Common digital forensic tools such as AccessData FTK Imager and Autopsy by Sleuth Kit Labs only show a narrow analysis of audio data information. Audio files are only shown one dimensionally in their native, text, or hex format. Both software also show the metadata of the file including date created, size, and file extension. There are no visual representations of audio files in these programs, meaning that a file embedded with a hidden message evade detection from these programs. Depending on the scenario of an investigation or case, this could mean life or death for civilians or a criminal roaming free within society. Therefore, it is important to consider all aspects of digital audio evidence as

hidden information can be undetectable with current forensic software.

IV. Methodology

The proposed system will investigate spectrogram capabilities for encoding hidden information and detecting unnatural anomalies. Developed in Python, the script will provide the user with a command-line interface to interact with. To capture the importance of spectrograms in digital audio forensics, there are three core characteristics the system must have, including:

- Spectrogram-based steganography,
- Audio reconstruction and embedding, and
- Forensic anomaly detection.

A. *Spectrogram-based Steganography*

As previously mentioned, spectrograms are a two-dimensional representation of audio, showcasing the frequency (pitch) of an audio over its duration. Loudness can also be measured as decibel information is also typically shown through color variances. Figure 1 shows an example of a spectrogram taken from an audio recording of a song.

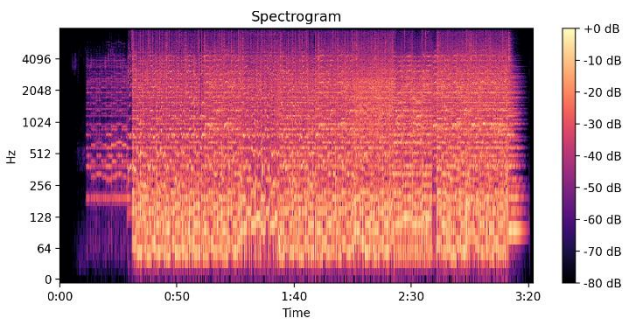

Fig. 1. Spectrogram of an audio recording of a song. Loudness is shown, where lighter colors represent louder frequencies.

In the script, users are prompted if they want to encode information in the form of an image or text into a spectrogram. This is a form of steganography, which is the practice of concealing information within other non-secret data. In the case of text encoding, the script generates a user-specified message into a grayscale image, where light pixels demonstrate letters and high magnitude, and dark pixels represent low magnitude and empty space. This image is processed, scale, and converted as a spectrogram.

If a user wants to embed images, the supplied image file will undergo a similar process. First, it is converted into grayscale and resized. This template image is used to convert the supplied image into a format adaptable to a spectrogram. The user is also free to change various embedding characteristics like font size, image size, and font selection.

B. *Audio Reconstruction and Embedding*

After the encoding process, the created spectrogram undergoes an audio reconstruction method using the Griffin-Lim algorithm. The Griffin-Lim algorithm is an iterative method that reconstructs a time-domain signal by estimating phase information and produces an audio signal that audibly resembles the visual structures produced by embedded content from the spectrogram. Optionally, the user is also able to instead embed the generated spectrogram into another audio file, which is a feasible scenario in digital forensic cases.

After producing a new audio signal, it then goes through minimal audio processing functions like normalization. This is used to prevent some manipulation detection by hiding clipping and insertion patterns. This process demonstrates spectrogram capabilities of audio manipulation through directly embedding information as frequencies within the audio file.

C. *Forensic Anomaly Detection*

To address the importance of spectrograms in anomaly detection, an inspection feature will be implemented to analyze audio files for possible spectrogram manipulation. The inspection process will compute a spectrogram of the chosen audio file. Focus will be placed on higher-frequency regions of the spectrogram as high frequency areas are typical for message visibility and less interference with the source audio. A binary mask of the spectrogram will be created to show areas of interest.

Statistical features of the spectrogram is then analyzed to classify if the audio file has been altered. Spectral density, which measures the proportion of high-energy (loud) areas, is analyzed to find natural audio and artificially embedded content. Row and column variance of the spectrogram, where intensity is measured and compared to neighboring frequencies, is also examined. Atypical structures will not exude a smooth transition of intensity like natural audio would. Finally, an edge score is calculated by measuring abrupt changes in magnitude between adjacent frequency ranges. Again, gradual changes would showcase natural audio, whereas sharp boundaries and edges can attribute to unnatural, artificial content.

V. Implementation & Evaluation

A. *Implementation*

The proposed methods were implemented in Python. Multiple scientific and audio processing libraries were utilized including NumPy, Librosa, Python Imaging Library (PIL), Matplotlib, and SciPy. Audio analysis and reconstruction were performed using Librosa, while spectrogram templates were generated using the Python Imaging Library (PIL). Other libraries were used for graph generation and calculations. Running the script through the terminal will prompt the user through a command line interface in which the user will interact with the software to access the various implemented features.

B. *Evaluation of Solution*

In order to test the solution, all characteristics of the system were tested and analyzed. First, the encoding capabilities were tested. When the user starts the script, they are prompted with

three modes: Text, Image, or Inspect. The Text mode allows the user to embed line of text into a spectrogram. Following the script, the user will choose the filename of the generated spectrogram, filename of the reconstructed audio file, and the message that they want to encode. Figure 2 shows the spectrogram of a message, “This is a hidden message in your spectrogram!” This spectrogram was not embedded on top of any existing audio file.

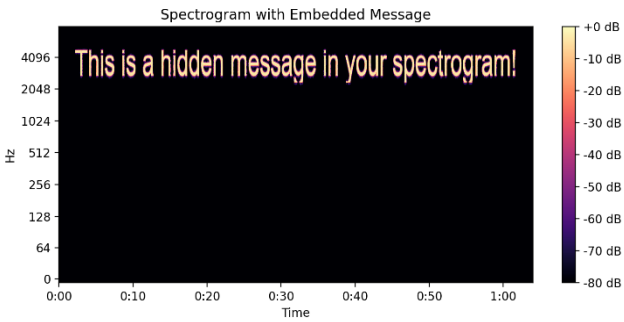

Fig. 2. Generated spectrogram from the user-specified message.

Next, the image embedding was tested with a photograph. When starting up the script, choosing the Image mode will also prompt the user to provide output filenames for spectrogram image and audio file. They will also be prompted to provide an image file as well. Figure 3 shows the image the application received and the output spectrogram the script produced.

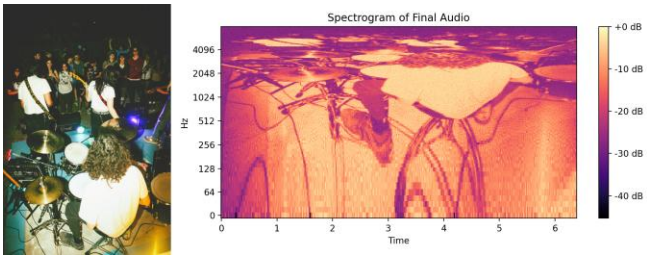

Fig. 3. Generated spectrogram (right) for a supplied image (left).

Though not as visible and clear as the text embedding, both spectrograms exhibit clear resemblance of their original content. Text were clear in their spectrograms, however images were limited to their native aspect ratio, often causing warping and shrinkage in the final generated spectrogram.

Audio reconstruction capabilities were also tested within this process. Because each generated spectrogram has an accompanying audio file, these audio files were able to examined in various forensic software. Analysis within FTK Imager and Autopsy showcase no indication of a secret embedded image or message as there are no audio visualization options like spectrogram analysis. Upon audible analysis, infrequent pitches and frequencies are hard, however no decipherable information can be observed from listening to the files as well.

Before the spectrogram process ends, users have the optional feature to instead embed the spectrogram over an existing audio file. This will hold both the original audio file’s spectrogram and the embedded content. This is a common case when trying to hide information in audio files as a song or voice recording is more inconspicuous than an audio recording of random pitches, like those generated from the spectrograms and just the embedded content. This feature was tested by embedding the previously mentioned hidden message into the audio file of a song. Figure 4 shows the resulting spectrogram when the user embeds a hidden message with an audio file of a song.

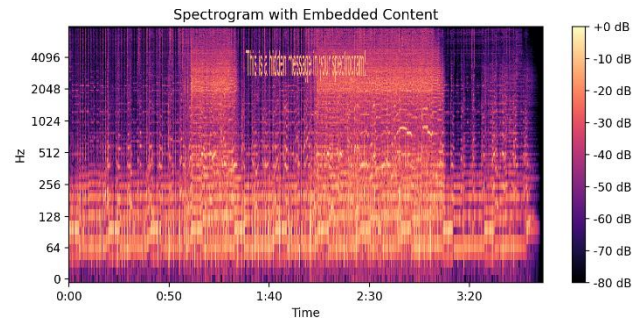

Fig. 4. Generated spectrogram from a hidden message merged with an audio file of a song

This result shows the hidden message within the original song’s frequencies. When listening to this audio file, the original song can be heard, and pitches corresponding to the hidden message will also be heard when it appears, for example, in Figure 4, random frequencies would be heard over the song around the 1 minute and 30 second mark.

Finally, detection capabilities were tested to see if the detection algorithm can classify possible audio files with embedding. Choosing the Inspect mode will prompt the user to choose an audio file to inspect. In this test, the audio file generated from the previous spectrogram in Figure 4 was used. The filename of the output spectrogram is also requested. Figure 5 showcases the output spectrogram of the inspected audio file and the flagged region that shows possible embedded content. It should be noted that because the audio file from Figure 4 was used, the same spectrogram was generated.

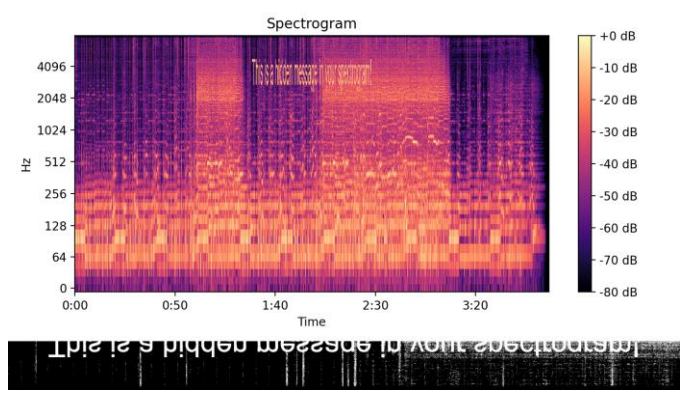

Fig. 5. Generated spectrogram from an inspected audio file (top) and the points of interest detected (bottom).

In this test, the algorithm was unable to fully detect the message, however it was able to mark half of it as possible points of interest. It also picked up some of the source audio's frequency information. This is due to it being a song for patterns of high noise, indicative of a song's chorus or various melodies of the song.

VI. Conclusion and Future Works

In conclusion, this paper examined spectrograms as an underutilized, yet significant layer in digital audio forensics, indicating that they can be used to embed and detect information explicitly through frequencies. A Python-based framework was developed to show that images and text can be manipulated and intentionally encoded into frequencies. These frequencies, when converted to a spectrogram, will reveal themselves to resemble hidden messages and content. These hidden messages can then be converted into audio files using the Griffin-Lim algorithm, which can act as unsuspecting audio files. Furthermore, these hidden messages can be encoded into existing audio files by merging the spectrograms together, which are harder to discern as malicious.

Examining these generated audio files in current forensic software will not showcase their visual representation. Only metadata and one dimensional formats of the audio files like hex and text are shown from the FTK Imager and Autopsy, two common digital forensic software. Without visual representations, like spectrograms, present in digital forensic software, these hidden messages were able to bypass traditional analysis methods like metadata inspection and waveform analysis.

To combat common digital audio analysis methods, a proposed methodology to investigate spectrograms was also implemented. Understanding the various features of spectrograms like spectral density, row and column variance, and edge detection can automate the analysis of spectrograms and further expedite evidence analysis.

Beyond demonstrating the feasibility of spectrograms in cyber security crimes and digital investigations, this work also contributed practical methodology designed to support the analysis of multimedia data. Data is not truly one sided, and understanding data in multiple spectrums can reveal information that is not readily available or discernable to the naked eye. Different layers hold information and mediums for users to exploit and manipulate to their own intent, so it is important to understand hidden outlets of file information, like spectrograms.

In the future, a lot of work can be done to iron out issues in the encoding process. For example, image encoding fell flat as it became entirely dependent on the image aspect ratio. Generated spectrograms would stretch or shrink areas of the image from the aspect ratio. Also, encoding hidden content into other audio files worked great for text, but would create a lot of unidentifiable content for images after spectrogram generation. Further testing the capabilities of audio encoding could employ further reliance on modern techniques.

Additionally, creating an easy-to-use interface would deem useful as well. This would help adapt the script into larger software while also indicating its usability. A command line interface is somewhat technical and inefficient for everyday use, and making it accessible would broaden the audience of users for the system.

Further adaptations can include artificial intelligence and machine learning integration. Utilizing modern techniques, anomaly detection capabilities are further enhanced. Currently, false positives can be picked up as a detected anomaly due to the nature of specific source audios. For example, music would have varying intensities and varying points in the song. Artificial intelligence and machine learning can be utilized to compare original source spectrograms with a generated spectrogram to extrapolate relevant, embedded content. With that, it can also store relevant content in multiple formats, all while being automatic.

In conclusion, this paper demonstrates that spectrograms are not only analytical and diagnostic tools, but also represent an important and exploitable layer of digital audio evidence. Recognizing and analyzing this layer enhances the forensic practitioner's ability to detect hidden manipulation, strengthens investigative efficiency, and contributes to the ongoing evolution of digital audio forensics in an investigative landscape.

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